

Chapter 7

Momentum

Gravity (2013)

- Science fiction thriller film directed by Alfonso Cuarón
- The story is about American astronauts who attempt to return to Earth after the destruction of their Space Shuttle in orbit.



Questions

Why is “you have to let me go” necessary to save the female astronaut?

What physics principle is involved in the sense?

Do you think the reaction force in the scene makes sense? (considering it happens in space.)

Introduction

- In many situations, such as a bullet hitting a carrot, we cannot use Newton's second law to solve problems because we know very little about the complicated forces involved.
- In this chapter, we shall introduce *momentum* and *impulse*, and the *conservation of momentum*, to solve such problems.

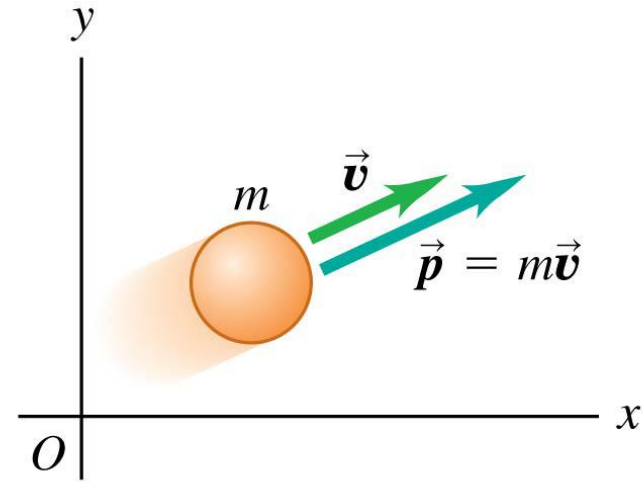


Momentum and Newton's second law

- The **momentum** of a particle is the product of its mass and its velocity:

$$\vec{p} = m\vec{v}$$

- Newton's second law can be written in terms of momentum:



Momentum $\vec{p}$ is a vector quantity; a particle's momentum has the same direction as its velocity $\vec{v}$.

Newton's second law

in terms of momentum:

The net force acting on a particle ...

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

... equals the rate of change of the particle's momentum.

Impulse

- The **impulse** of a force is the product of the force and the time interval during which it acts:

Impulse of a constant net force $\rightarrow \vec{I} = \sum \vec{F}(t_2 - t_1) = \sum \vec{F} \Delta t$

Constant net force

Time interval over which net force acts

- On a graph of $\sum F_x$ versus time, the impulse is equal to the area under the curve:

Impulse of a general net force (either constant or varying) $\rightarrow \vec{I} = \int_{t_1}^{t_2} \sum \vec{F} dt$

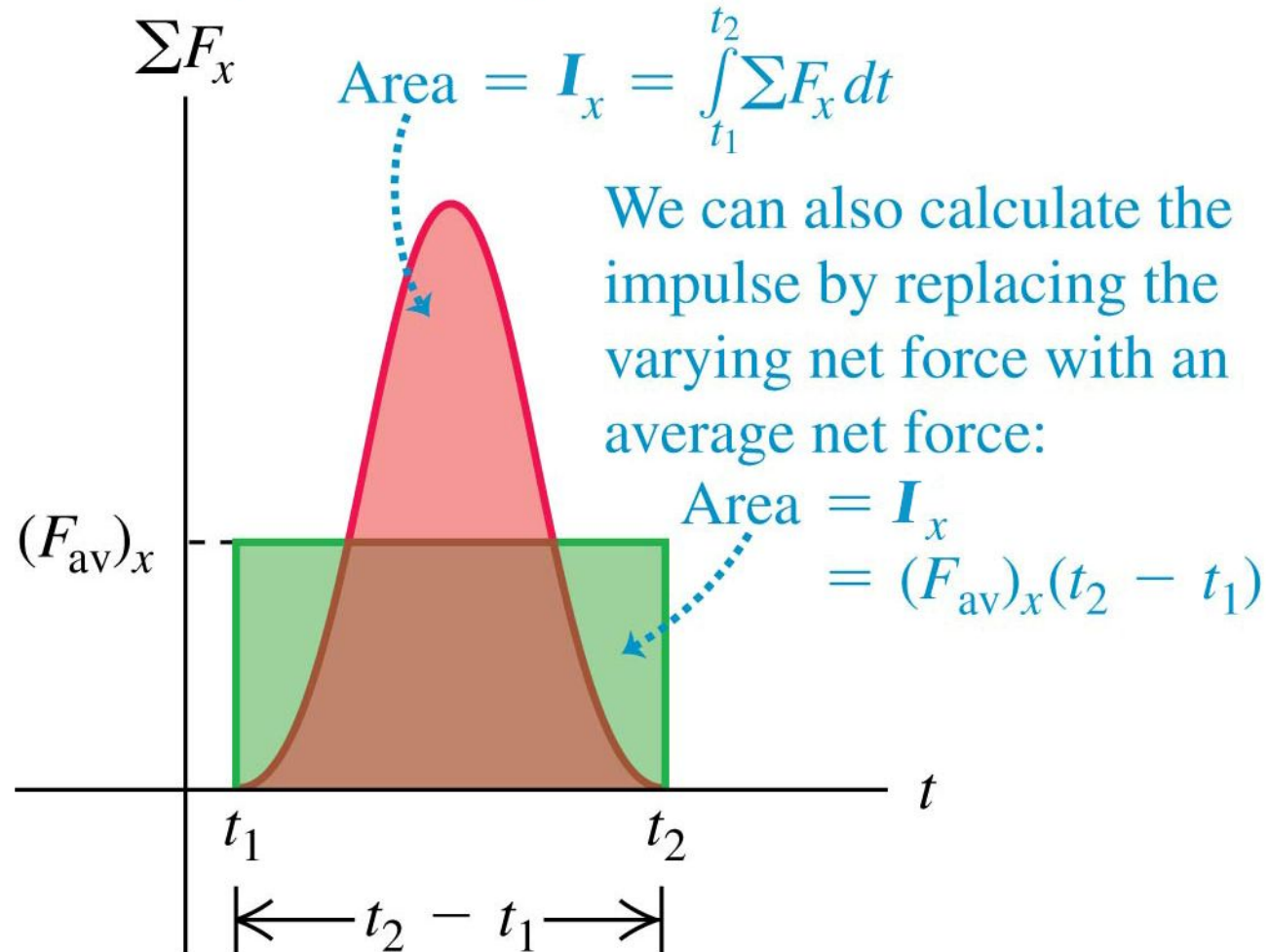
Upper limit = final time

Time integral of net force

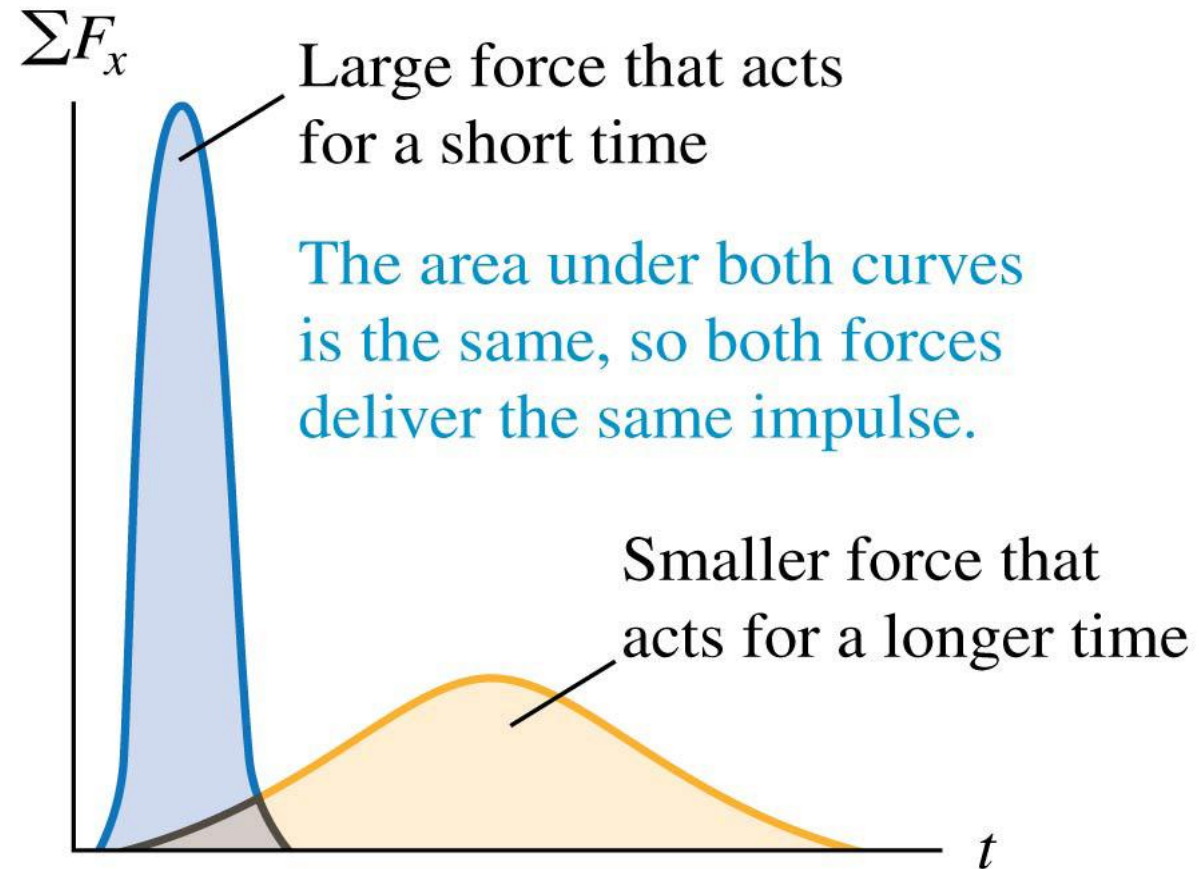
Lower limit = initial time

Impulse

The area under the curve of net force versus time equals the impulse of the net force:



Impulse



Impulse and momentum

- **Impulse–momentum theorem:** The change in momentum of a particle during a time interval is equal to the impulse of the net force acting on the particle during that interval:

The diagram shows the impulse-momentum theorem equation $\vec{I} = \vec{p}_2 - \vec{p}_1 = \Delta\vec{p}$ on a light yellow background. Four blue dotted arrows point from descriptive text labels to parts of the equation: from 'Impulse of net force over a time interval' to $\vec{I}$, from 'Final momentum' to $\vec{p}_2$, from 'Initial momentum' to $\vec{p}_1$, and from 'Change in momentum' to $\Delta\vec{p}$.

$$\vec{I} = \vec{p}_2 - \vec{p}_1 = \Delta\vec{p}$$

Labels and arrows:

- Impulse of net force over a time interval points to $\vec{I}$.
- Final momentum points to $\vec{p}_2$.
- Initial momentum points to $\vec{p}_1$.
- Change in momentum points to $\Delta\vec{p}$.

Quiz

You are testing a new car using crash test dummies. Consider two ways to slow the car from 90 km/h to a complete stop:

- (i) You let the car slam into a wall, bringing it to a sudden stop.
- (ii) You let the car plow into a giant tub of gelatin so that it comes to a gradual halt.

In which case is there a greater *impulse* of the net force on the car?

- A. in case (i)
- B. in case (ii)
- C. The impulse is the same in both cases.
- D. not enough information given to decide

Quiz

A ping-pong ball and a bowling ball have the same momentum. Which one has the larger kinetic energy?

- (a) the ping-pong ball
- (b) the bowling ball
- (c) They have the same kinetic energy.

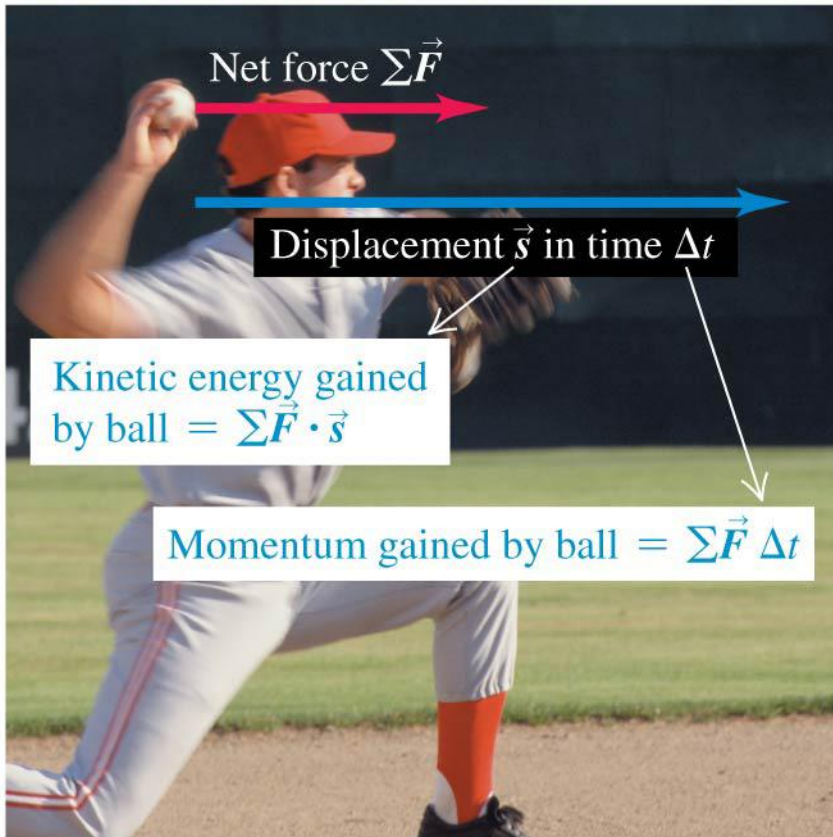
Impulse and momentum

- When you land after jumping upward, your momentum changes from a downward value to zero.
- It's best to land with your knees bent so that your legs can flex.
- You then take a relatively long time to stop, and the force that the ground exerts on your legs is small.
- If you land with your legs extended, you stop in a short time, the force on your legs is larger, and the possibility of injury is greater.



Compare momentum and kinetic energy

- The kinetic energy of a pitched baseball is equal to the work the pitcher does on it (force multiplied by the distance the ball moves during the throw).
- The momentum of the ball is equal to the impulse the pitcher imparts to it (force multiplied by the time it took to bring the ball up to speed).



Principle of linear impulse and momentum

We obtain the principle of impulse and momentum by integrating the equation of motion with respect to time.

It can be applied to problems involving both linear and angular motion.

This principle is useful for solving problems that involve force, velocity, and time.

Principle of linear impulse and momentum

The principle of linear impulse and momentum is obtained by integrating the equation of motion with respect to time. The equation of motion can be written

$$\sum \mathbf{F} = m \mathbf{a} = m (d\mathbf{v}/dt)$$

Separating variables and integrating between the limits $\mathbf{v} = \mathbf{v}_1$ at $t = t_1$ and $\mathbf{v} = \mathbf{v}_2$ at $t = t_2$ results in

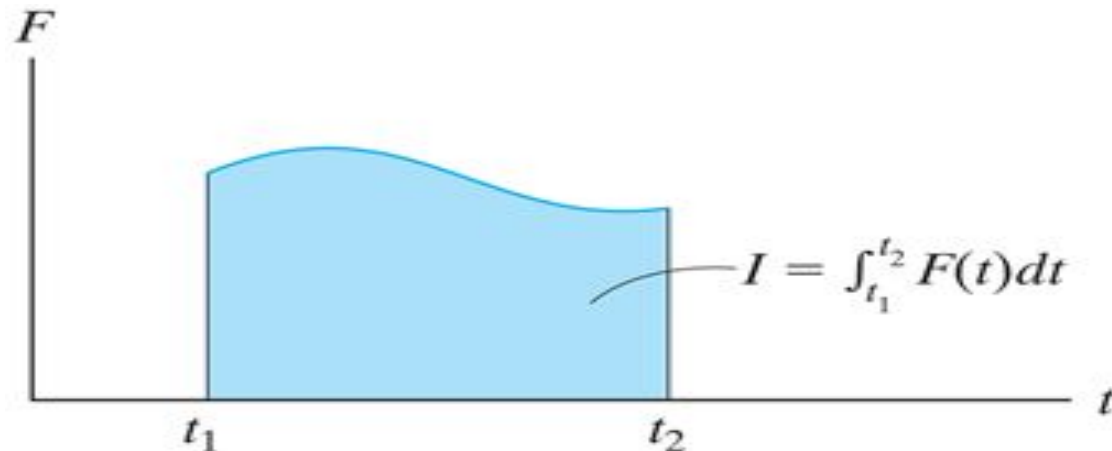
$$\sum \int_{t_1}^{t_2} \mathbf{F} dt = m \int_{\mathbf{v}_1}^{\mathbf{v}_2} d\mathbf{v} = m\mathbf{v}_2 - m\mathbf{v}_1$$

This equation represents the principle of linear impulse and momentum. It relates the particle's final velocity ($\mathbf{v}_2$) and initial velocity ($\mathbf{v}_1$) and the forces acting on the particle as a function of time.

Principle of linear impulse and momentum

Linear momentum: The vector $m\mathbf{v}$ is called the linear momentum, denoted as $\mathbf{P}$. This vector has the same direction as $\mathbf{v}$. The linear momentum vector has units of $(\text{kg}\cdot\text{m})/\text{s}$.

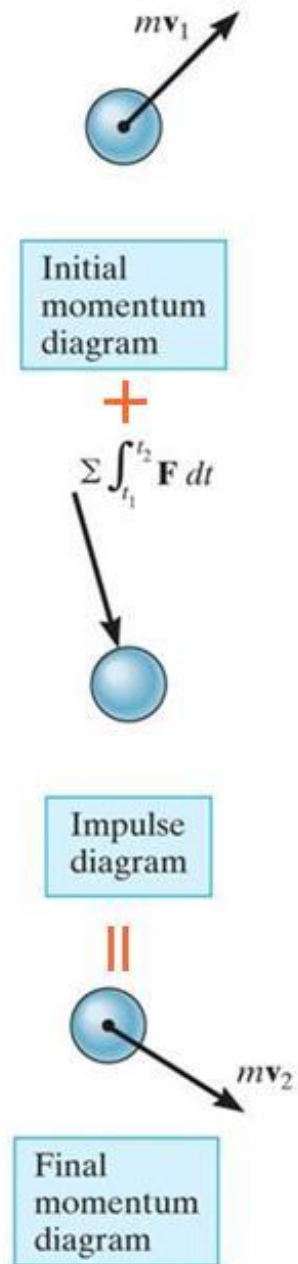
Linear impulse: The integral $\int \mathbf{F} dt$ is the linear impulse, denoted $\mathbf{I}$. It is a vector quantity measuring the effect of a force during its time interval of action. $\mathbf{I}$ acts in the same direction as $\mathbf{F}$ and has units of $\text{N}\cdot\text{s}$.



The impulse may be determined by direct integration. Graphically, it can be represented by the area under the force versus time curve. If F is constant, then

$$\mathbf{I} = \mathbf{F} (t_2 - t_1) .$$

Principle of linear impulse and momentum



The principle of linear impulse and momentum in **vector** form is written as

$$m\mathbf{v}_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$$

The particle's initial momentum plus the sum of all the impulses applied from t_1 to t_2 is equal to the particle's final momentum.

The two **momentum diagrams** indicate direction and magnitude of the particle's initial and final momentum, $m\mathbf{v}_1$ and $m\mathbf{v}_2$. The **impulse diagram** is similar to a free-body diagram, but includes the time duration of the forces acting on the particle.

Impulse and momentum: scalar equations

Since the principle of linear impulse and momentum is a vector equation, it can be resolved into its x, y, z component

scalar equations:

$$\begin{aligned} m(v_x)_1 + \sum \int_{t_1}^{t_2} F_x dt &= m(v_x)_2 \\ m(v_y)_1 + \sum \int_{t_1}^{t_2} F_y dt &= m(v_y)_2 \\ m(v_z)_1 + \sum \int_{t_1}^{t_2} F_z dt &= m(v_z)_2 \end{aligned}$$

The scalar equations provide a convenient means for applying the principle of linear impulse and momentum once the velocity and force vectors have been resolved into x, y, z components.

Problem solving

- Establish the x, y, z coordinate system.
- Draw the particle's free-body diagram and establish the direction of the particle's initial and final velocities, drawing the impulse and momentum diagrams for the particle. Show the linear momenta and force impulse vectors.
- Resolve the force and velocity (or impulse and momentum) vectors into their x, y, z components, and apply the principle of linear impulse and momentum using its scalar form.
- Forces as functions of time must be integrated to obtain impulses. If a force is constant, its impulse is the product of the force's magnitude and time interval over which it acts.

Quiz

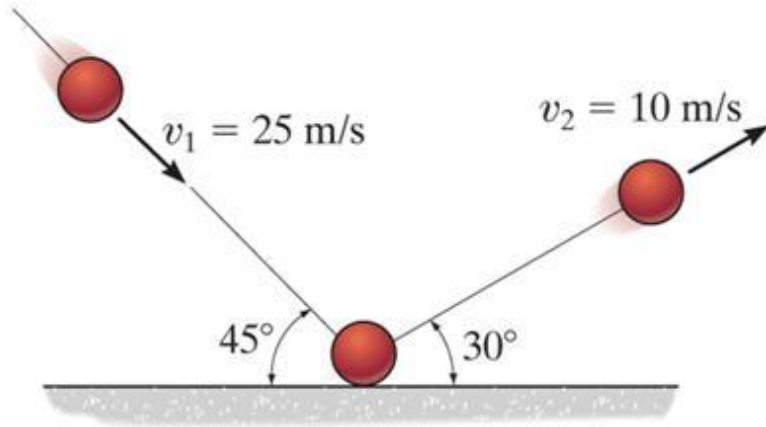
1. The linear impulse and momentum equation is obtained by integrating the _____ with respect to time.

- A) friction force B) equation of motion
- C) kinetic energy D) potential energy

2. Which parameter is not involved in the linear impulse and momentum equation?

- A) Velocity B) Displacement
- C) Time D) Force

Example



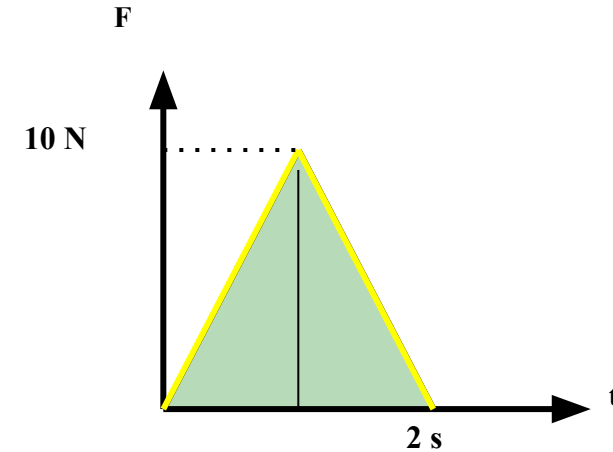
Given: A 0.5-kg ball strikes the rough ground and rebounds with the velocities shown. Neglect the ball's weight during the time it impacts the ground.

Find: The magnitude of the impulse exerted on the ball.

Quiz

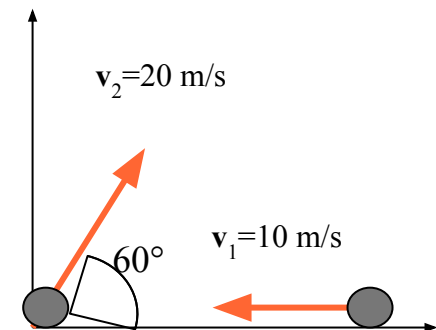
1. Calculate the impulse due to the force.

- A) $20 \text{ kg}\cdot\text{m/s}$ B) $10 \text{ kg}\cdot\text{m/s}$
C) $5 \text{ N}\cdot\text{s}$ D) $15 \text{ N}\cdot\text{s}$

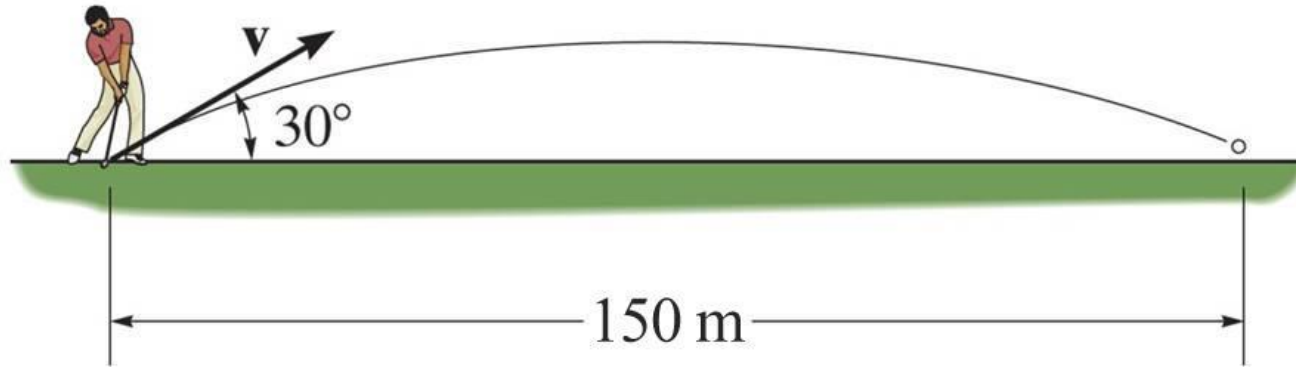


2. A constant force $\mathbf{F}$ is applied for 2 s to change the particle's velocity from $\mathbf{v}_1$ to $\mathbf{v}_2$. Determine the force $\mathbf{F}$ if the particle's mass is 2 kg.

- A) $(17.3 \mathbf{j}) \text{ N}$ B) $(-10 \mathbf{i} + 17.3 \mathbf{j}) \text{ N}$
C) $(20 \mathbf{i} + 17.3 \mathbf{j}) \text{ N}$ D) $(10 \mathbf{i} + 17.3 \mathbf{j}) \text{ N}$



Example



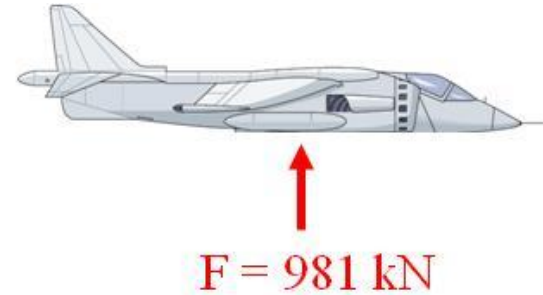
Given: A 0.05-kg golf ball is struck by the club and travels along the trajectory shown, $\theta = 30^\circ$, $R = 150$ m. Assume the club maintains contact with the ball for 0.5 ms.

Find: The average impulsive force exerted on the ball.

Quiz

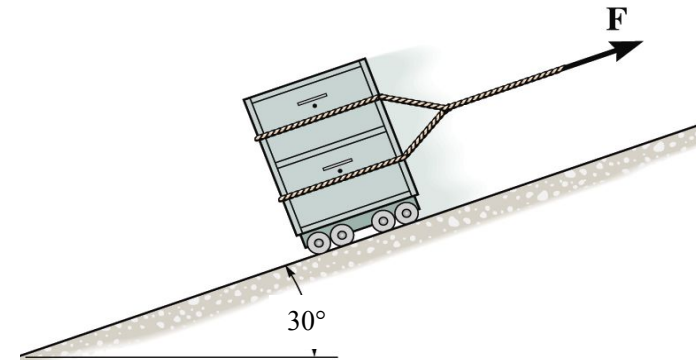
1. Jet engines on the 100-Mg VTOL aircraft exert a constant vertical force of 981 kN as it hovers. Determine the net impulse on the aircraft over $t = 10$ s.

- A) $-981 \text{ kN}\cdot\text{s}$ B) $0 \text{ kN}\cdot\text{s}$
C) $981 \text{ kN}\cdot\text{s}$ D) $9810 \text{ kN}\cdot\text{s}$



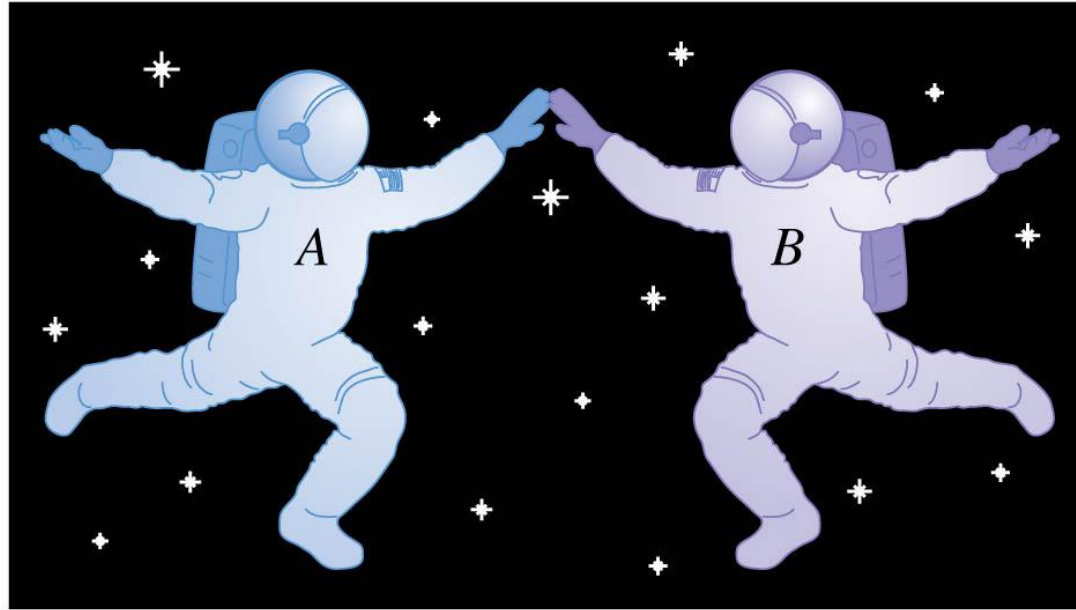
2. A 100-N(≈ 10 -kg) cabinet is placed on a smooth surface. If a force of a 100 N is applied for 2 s, determine the magnitude of the net impulse on the cabinet during this time interval.

- A) $0 \text{ N}\cdot\text{s}$ B) $100 \text{ N}\cdot\text{s}$
C) $200 \text{ N}\cdot\text{s}$ D) $300 \text{ N}\cdot\text{s}$

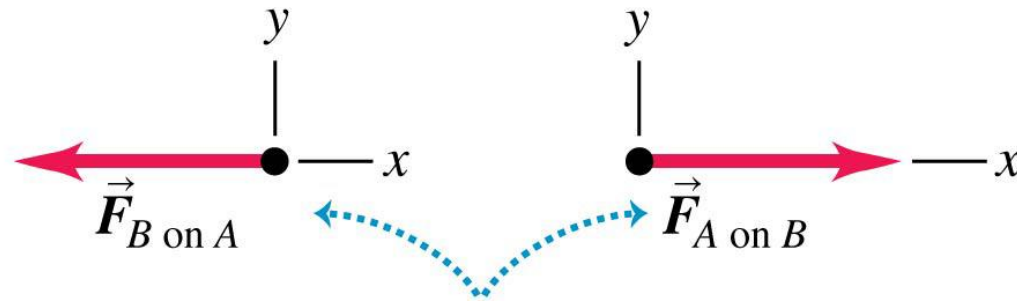


An isolated system

- Two astronauts push each other as they float freely in the zero-gravity environment of space.
- There are *no* external forces; when this is the case, we have an **isolated system**.



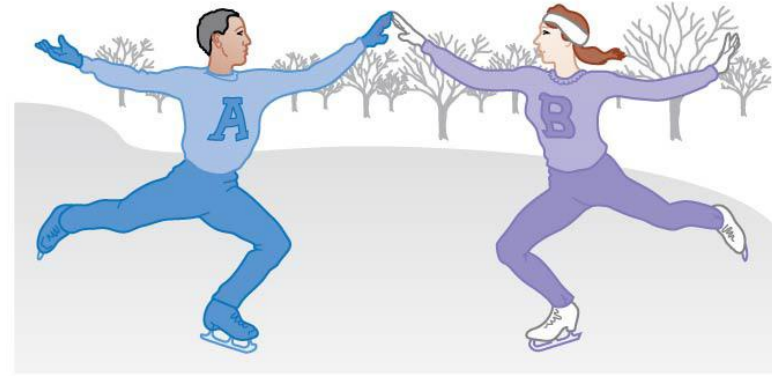
No external forces act on the two-astronaut system, so its total momentum is conserved.



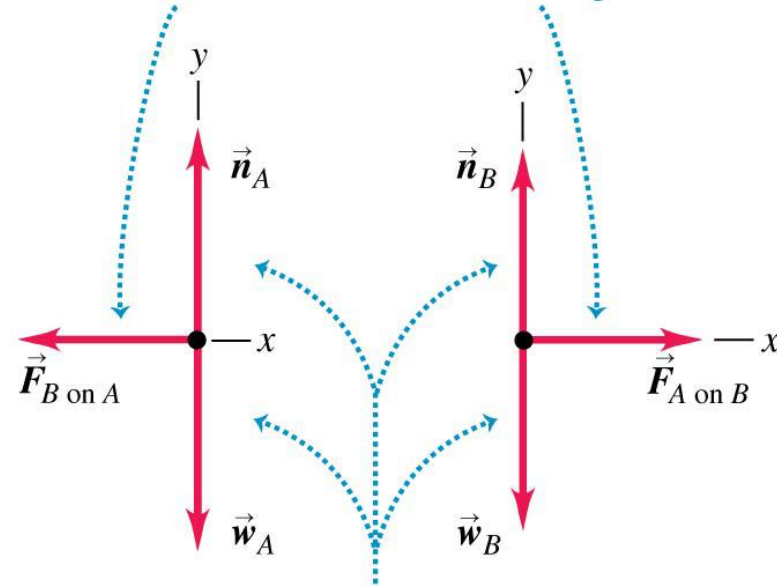
The forces the astronauts exert on each other form an action–reaction pair.

Conservation of momentum

- External forces (the normal force and gravity) act on the skaters shown, but their vector sum is zero.
- Therefore the total momentum of the skaters is conserved.
- **Conservation of momentum:** If the vector sum of the external forces on a system is zero, the total momentum of the system is constant.



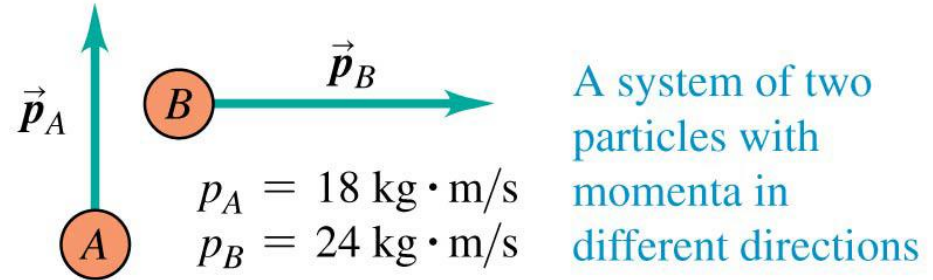
The forces the skaters exert on each other form an action–reaction pair.



Although the normal and gravitational forces are external, their vector sum is zero, so the total momentum is conserved.

Remember that momentum is a vector!

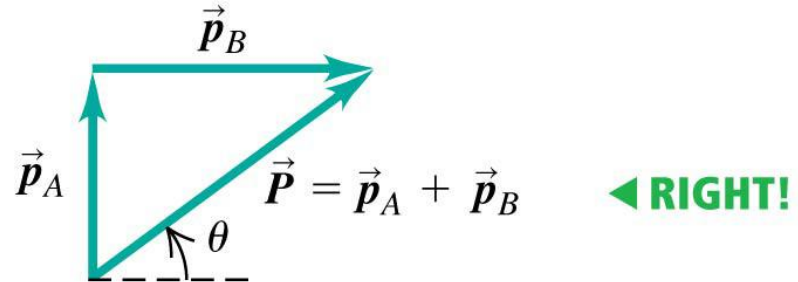
- When applying conservation of momentum, remember that momentum is a vector quantity!
- Use vector addition to add momenta, as shown at the right.



You CANNOT find the magnitude of the total momentum by adding the magnitudes of the individual momenta!

$$P = p_A + p_B = 42 \text{ kg} \cdot \text{m/s} \quad \text{◀ WRONG}$$

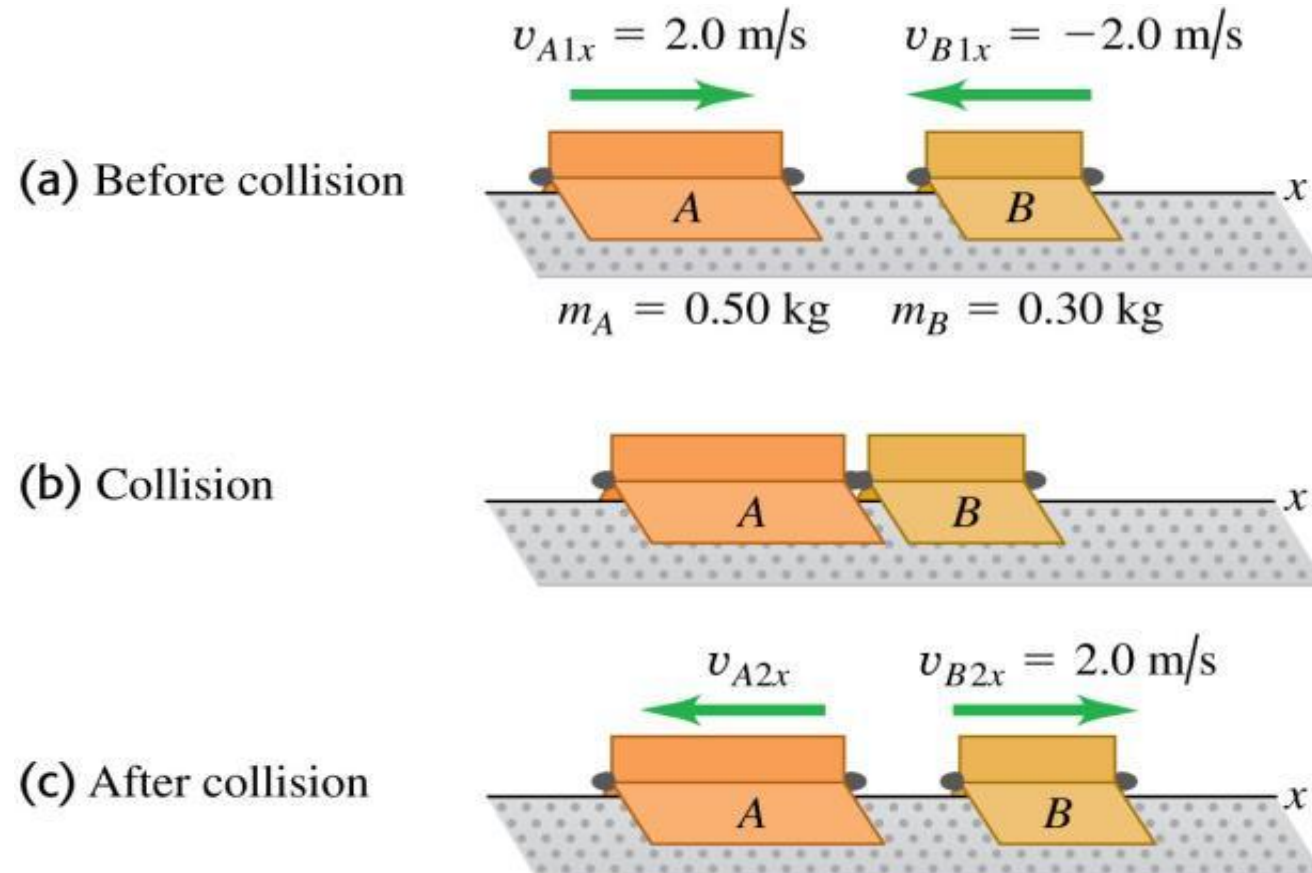
Instead, use vector addition:



$$\begin{aligned} P &= |\vec{p}_A + \vec{p}_B| \\ &= 30 \text{ kg} \cdot \text{m/s} \text{ at } \theta = 37^\circ \end{aligned}$$

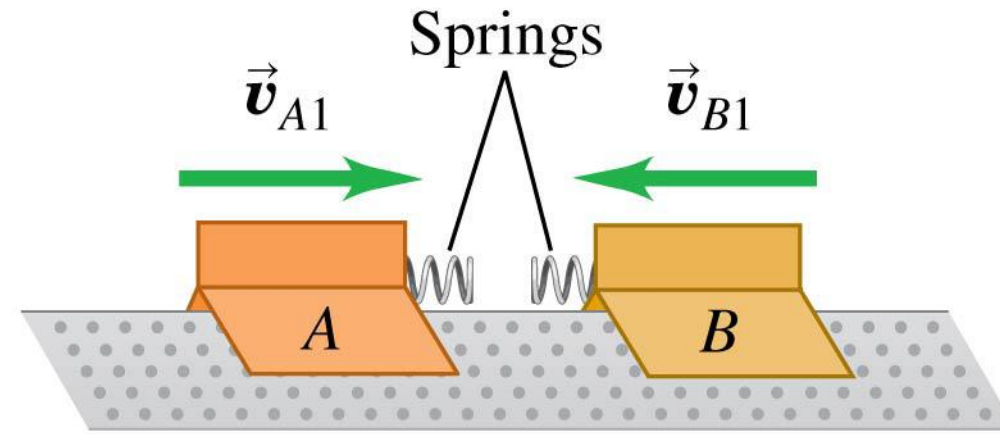
Objects colliding along a straight line

- Two gliders collide on an air track.
- We can find v_{A2x} and the changes in momenta and velocities of the gliders.



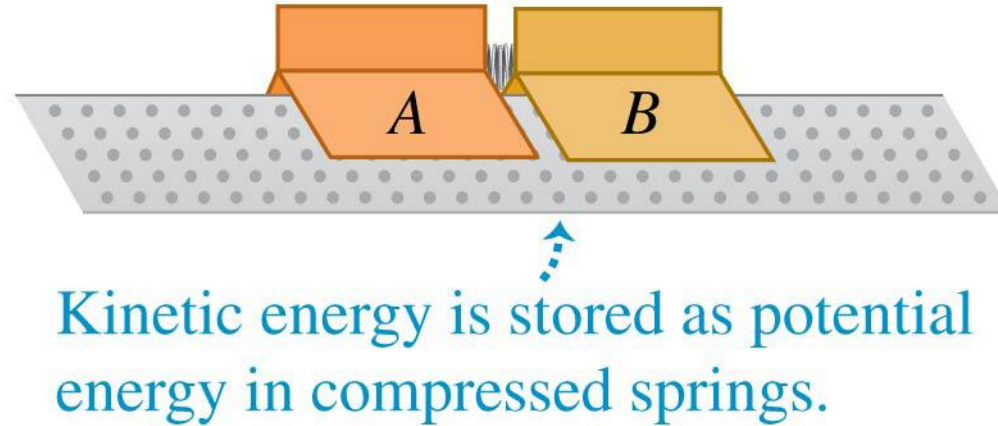
Elastic collisions: Before

- In an *elastic collision*, the total kinetic energy of the system is the same after the collision as before.



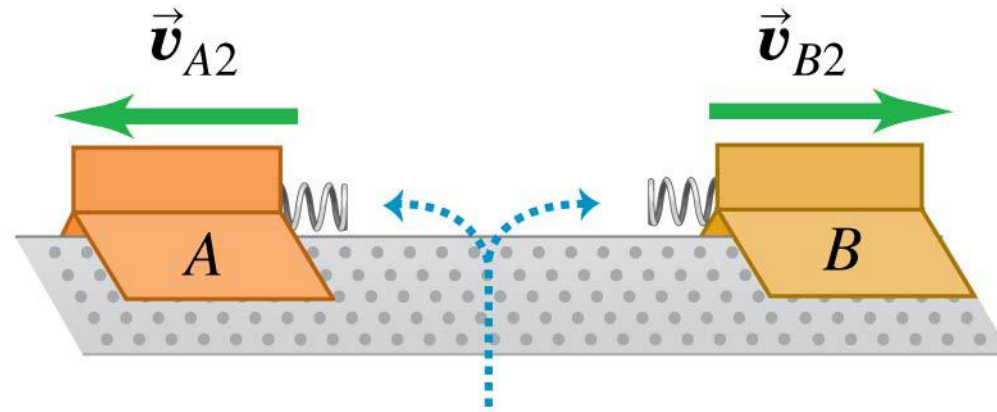
Elastic collisions: During

- In an *elastic collision*, the total kinetic energy of the system is the same after the collision as before.



Elastic collisions: After

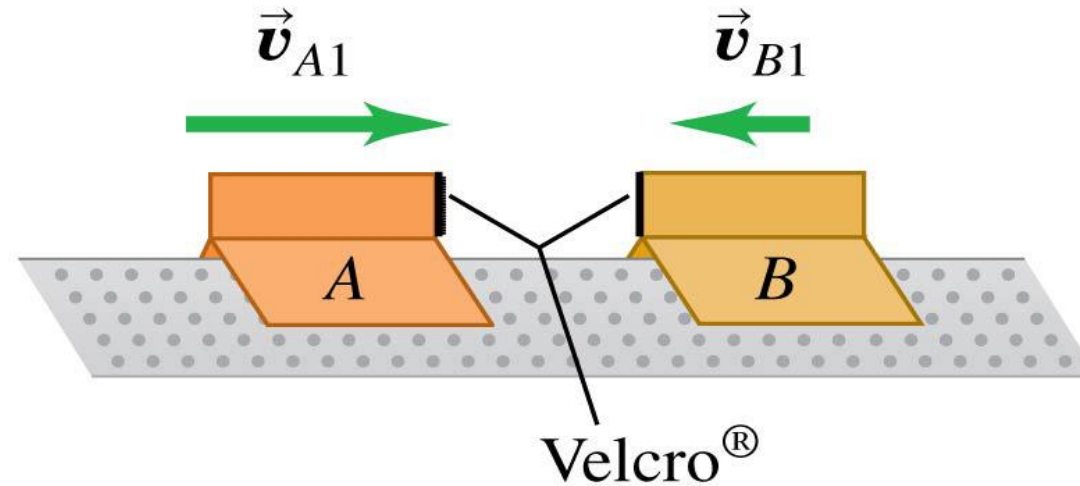
- In an *elastic collision*, the total kinetic energy of the system is the same after the collision as before.



The system of the two gliders has the same kinetic energy after the collision as before it.

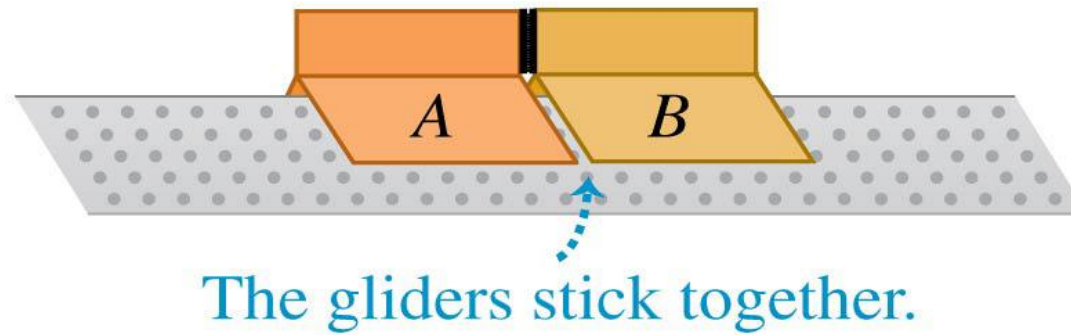
Completely inelastic collisions: Before

- In an *inelastic collision*, the total kinetic energy after the collision is less than before the collision.



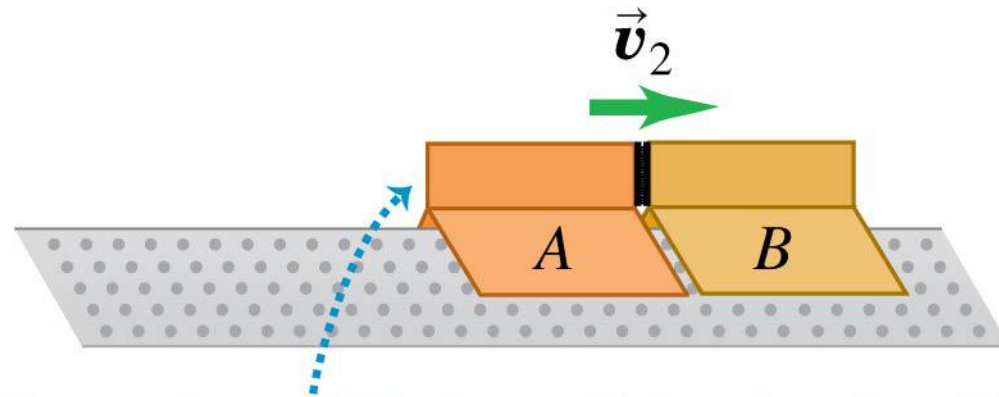
Completely inelastic collisions: During

- In an *inelastic collision*, the total kinetic energy after the collision is less than before the collision.



Completely inelastic collisions: After

- In an *inelastic collision*, the total kinetic energy after the collision is less than before the collision.



The system of the two gliders has less kinetic energy after the collision than before it.

Collisions

- In an *inelastic collision*, the total kinetic energy after the collision is less than before the collision.
- A collision in which the bodies stick together is called a **completely inelastic collision**.
- In *any* collision in which the external forces can be neglected, the total momentum is conserved.
- In elastic collisions *only*, the total kinetic energy before equals the total kinetic energy after.

Some inelastic collisions



- Cars are intended to have inelastic collisions so the car absorbs as much energy as possible.
- The crumple zone (crush space) in an automobile absorbs energy during a collision by controlled deformation.

- This absorbed energy cannot be recovered, since it goes into a permanent deformation of the car.



Elastic collision example

- Billiard balls deform very little when they collide, and they quickly spring back from any deformation they do undergo.



- Hence the force of interaction between the balls is almost perfectly conservative, and the collision is almost perfectly elastic.

Elastic collisions in one dimension

- Let's look at a one-dimensional elastic collision between two bodies A and B , in which all the velocities lie along the same line.
- We will concentrate on the particular case in which body B is at rest before the collision.
- Conservation of kinetic energy and momentum give the result for the final velocities of A and B

Elastic collisions

- For a collision of two bodies, consider the line of action between them

- $$\left\{ \begin{array}{l} m_A \vec{v}_{A1} + m_B \vec{v}_{B1} = m_A \vec{v}_{A2} + m_B \vec{v}_{B2} \\ \frac{1}{2} m_A \vec{v}_{A1}^2 + \frac{1}{2} m_B \vec{v}_{B1}^2 = \frac{1}{2} m_A \vec{v}_{A2}^2 + \frac{1}{2} m_B \vec{v}_{B2}^2 \end{array} \right.$$

If body B is initially at rest (i.e. $v_{B1} = 0$), solving the above gives

$$\vec{v}_{A2} = \frac{m_A - m_B}{m_A + m_B} \vec{v}_{A1}$$

$$\vec{v}_{B2} = \frac{2m_A}{m_A + m_B} \vec{v}_{A1}$$

- Consider the cases where $m_A = m_B$, $m_A \gg m_B$ and $m_A \ll m_B$

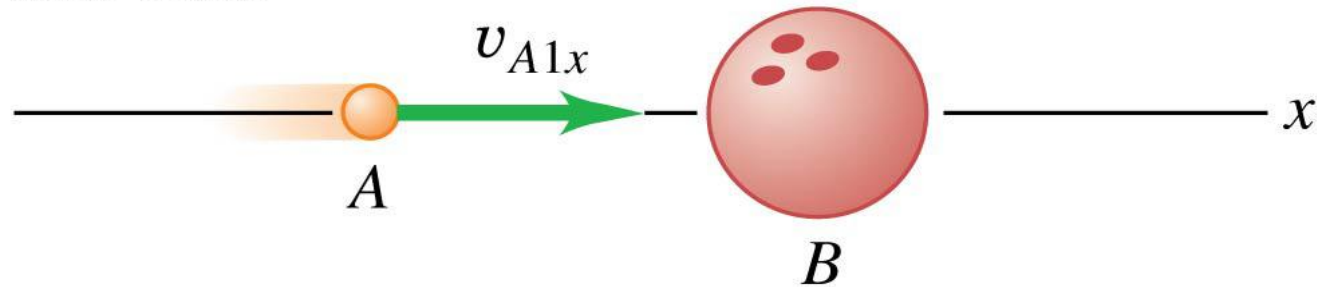
Elastic collisions (with B initially at rest)

Elastic collisions (with B initially at rest)

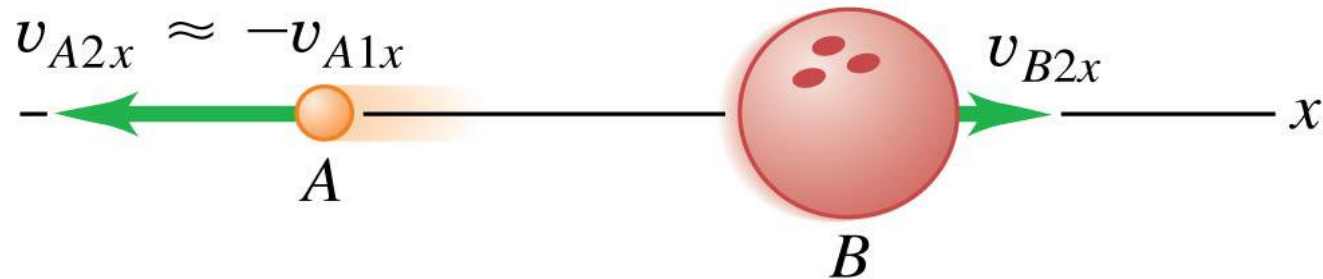
Elastic collisions in one dimension

- When B is much more massive than A , then A reverses its velocity direction, and B hardly moves.

BEFORE

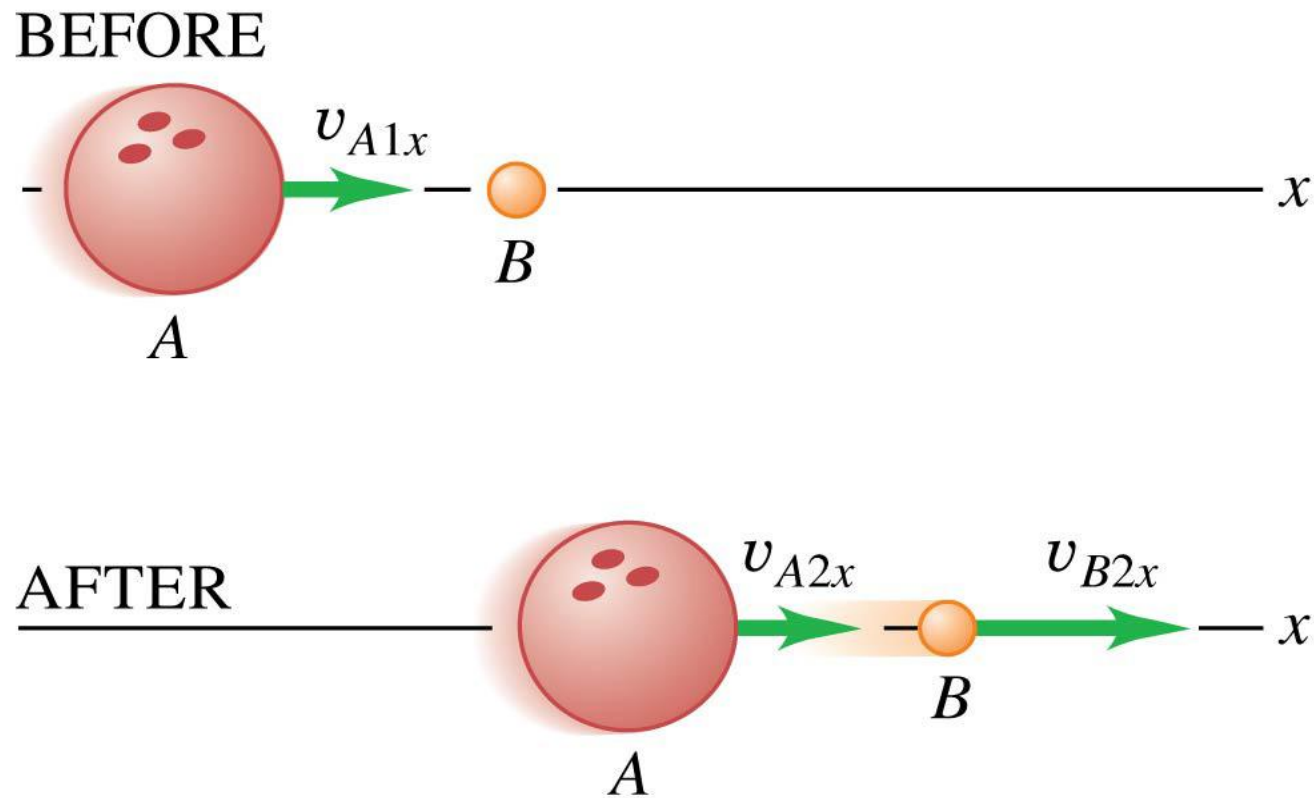


AFTER



Elastic collisions in one dimension

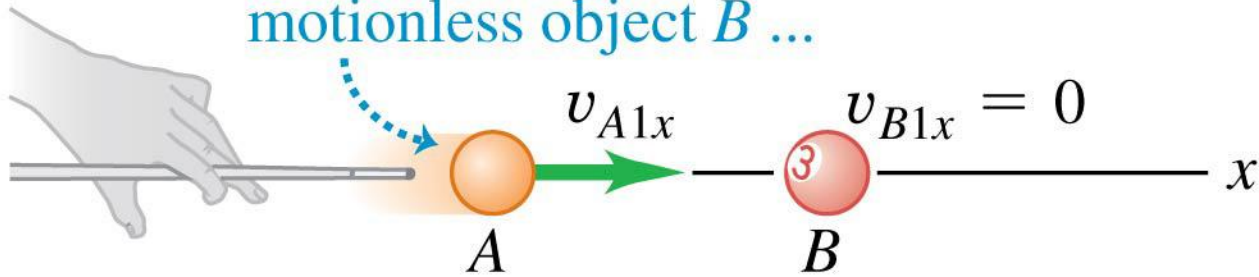
- When B is much less massive than A , then A slows a little bit, while B picks up a velocity of about twice the original velocity of A .



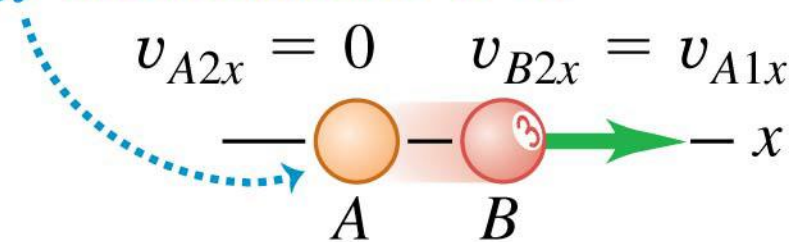
Elastic collisions in one dimension

- When A and B have equal masses, then A stops after the collision and B moves with the original speed of A .

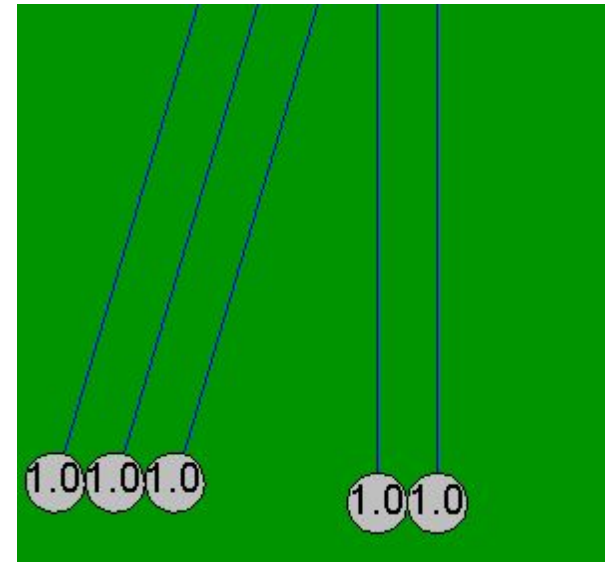
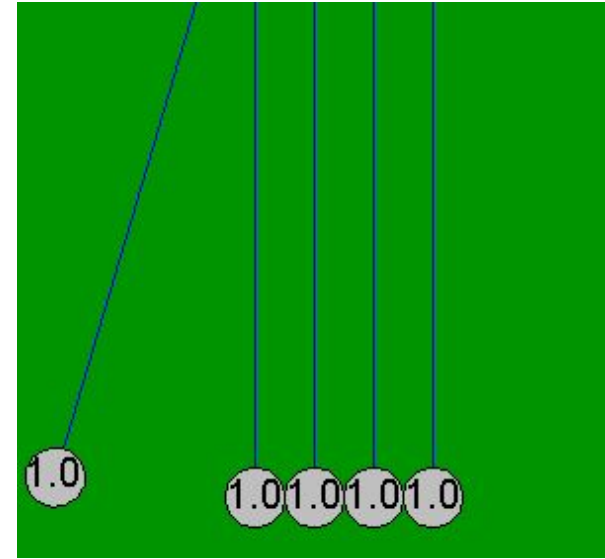
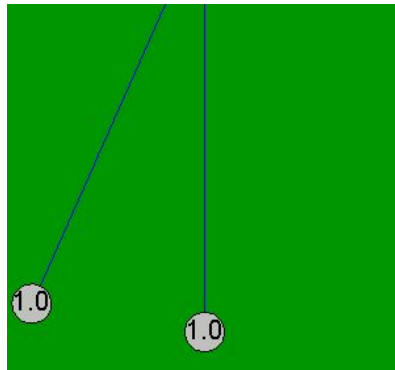
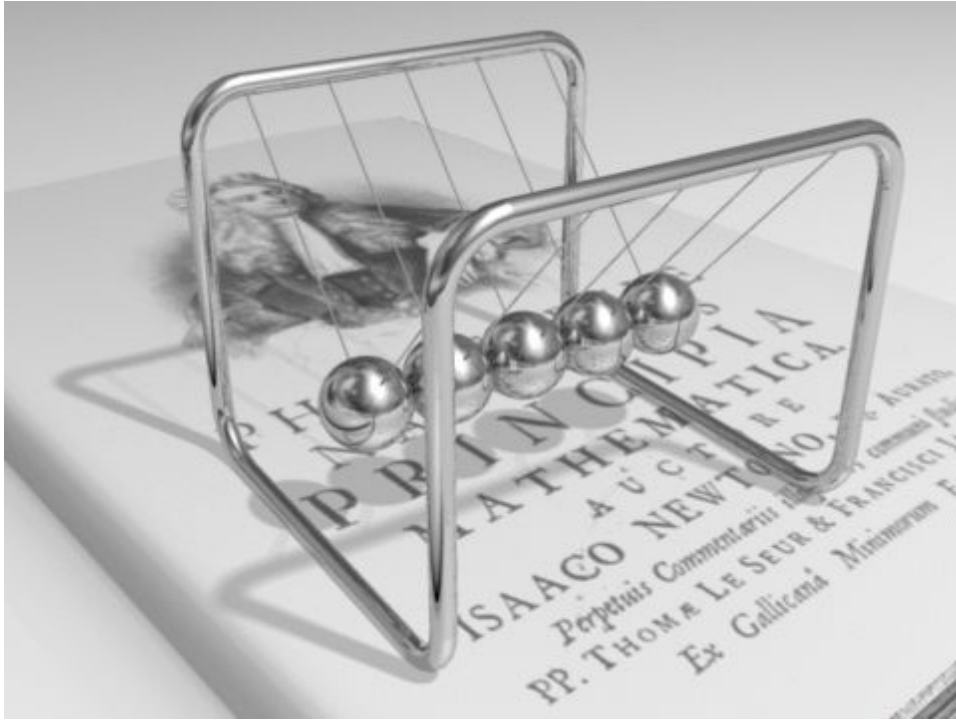
When a moving object A has a 1-D elastic collision with an equal-mass, motionless object B ...



... all of A 's momentum and kinetic energy are transferred to B .



Newton's cradle



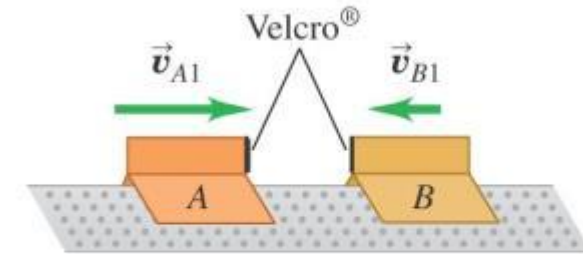
Elastic collisions in 1d (general case)

Elastic collisions in 1d (general case)

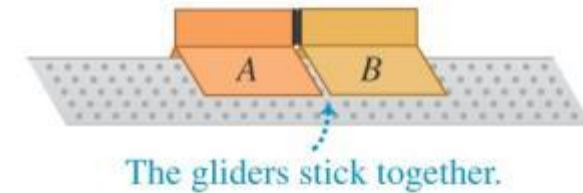
Inelastic collisions

- In an *inelastic collision*, the total kinetic energy after the collision is **less than before** the collision.
- A collision in which the bodies stick together is called a *completely inelastic collision* (see Figure at the right).
- In *any* collision in which the external forces can be neglected, the total momentum is conserved.

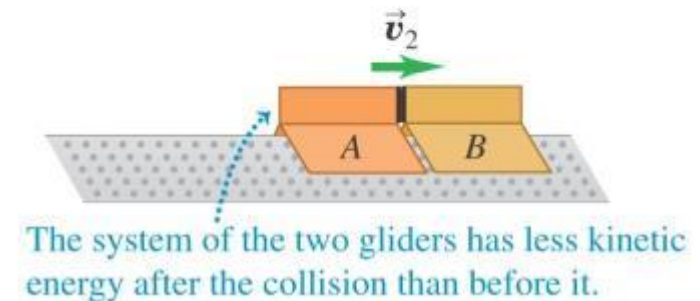
(a) Before collision



(b) Completely inelastic collision



(c) After collision



Inelastic collisions

- Conservation of momentum

$$m_A \vec{v}_{A1} + m_B \vec{v}_{B1} = (m_A + m_B) \vec{v}_2$$

- If m_B is initially at rest, i.e. $v_{B1} = 0$

$$\vec{v}_2 = \frac{m_A}{m_A + m_B} \vec{v}_{A1}$$

- Initial and final kinetic energies are

$$KE_1 = \frac{1}{2} m_A \vec{v}_{A1}^2$$

$$KE_2 = \frac{1}{2} (m_A + m_B) \vec{v}_2^2 = \frac{1}{2} \frac{m_A^2}{m_A + m_B} \vec{v}_{A1}^2$$

$$\boxed{\frac{KE_2}{KE_1} = \frac{m_A}{m_A + m_B}}$$

Center of mass

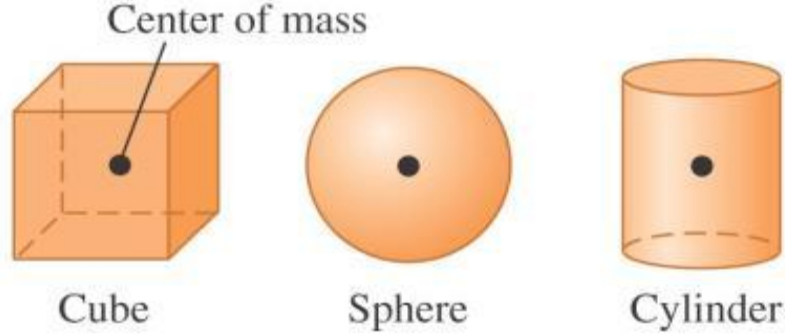
- We can restate the principle of conservation of momentum in a useful way by using the concept of center of mass.
- Suppose we have several particles with masses m_1 , m_2 , and so on.
- We define the center of mass of the system as the point at the position given by:

Position vector of center of mass of a system of particles $\vec{r}_{cm}$ = $\frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots}$ = $\frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$

Position vectors of individual particles

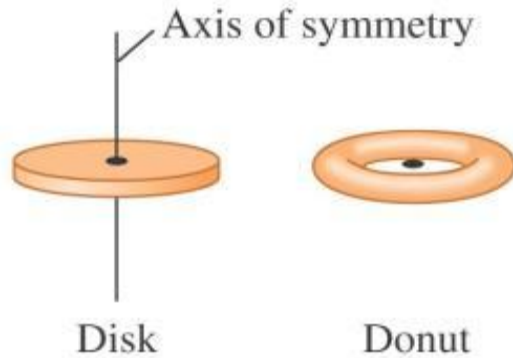
Masses of individual particles

Center of mass of symmetrical objects



If a homogeneous object has a geometric center, that is where the center of mass is located.

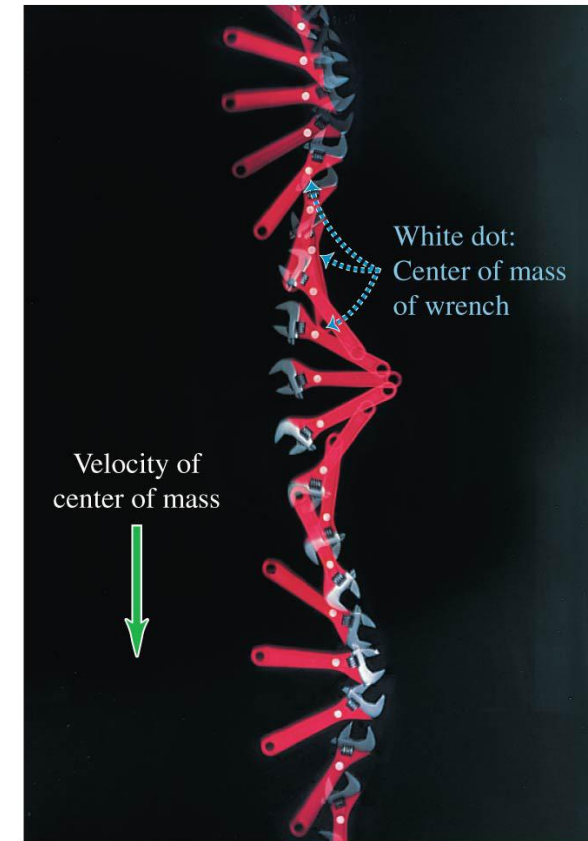
- It is easy to find the center of mass of a homogeneous symmetric object, as shown here.



If an object has an axis of symmetry, the center of mass lies along it. As in the case of the donut, the center of mass may not be within the object.

Motion of the center of mass

- The total momentum of a system is equal to the total mass times the velocity of the center of mass.
- The center of mass of the wrench at the right moves as though all the mass were concentrated there.



Total mass of a
system of particles

Momenta of individual particles

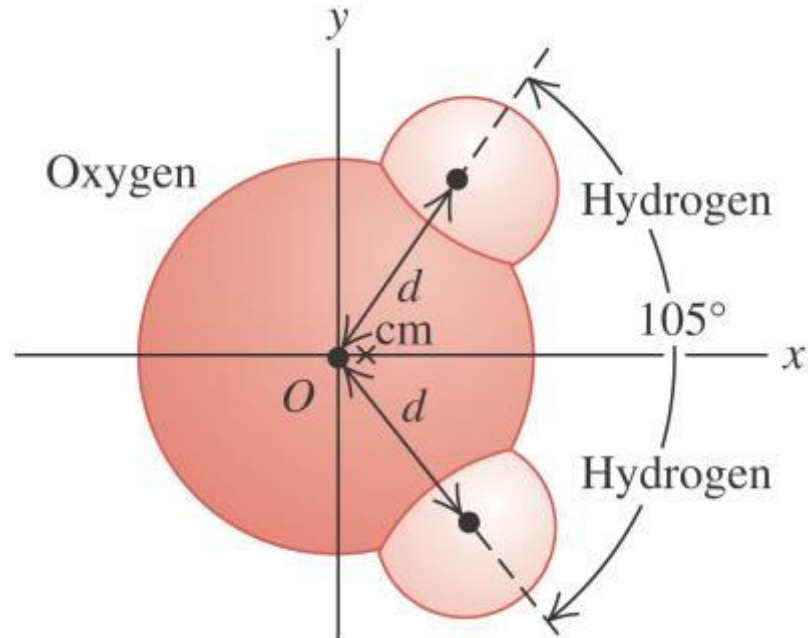
Velocity of
center of mass

$$M\vec{v}_{\text{cm}} = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 + \dots = \vec{P}$$

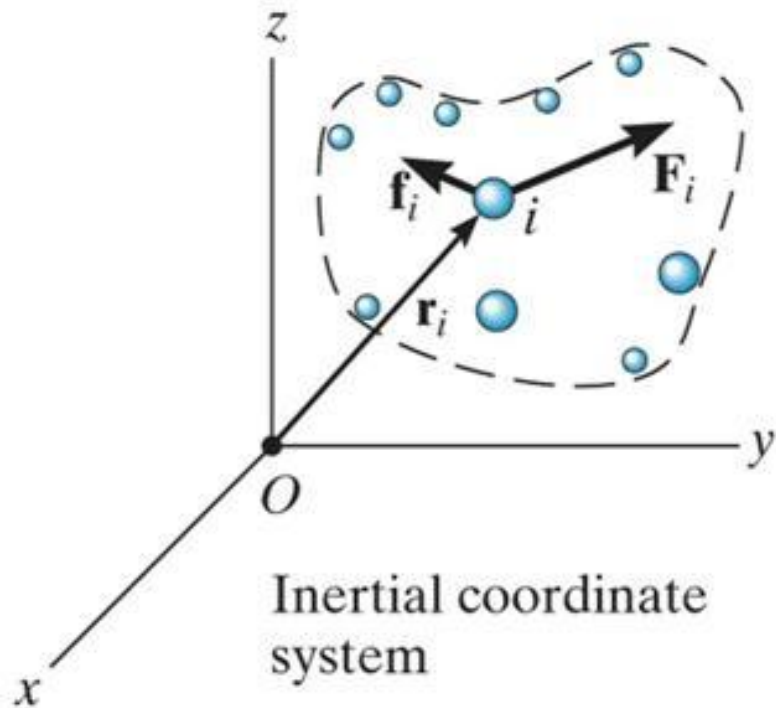
Total momentum of system

Center of mass of a water molecule

Figure 8.27 shows a simple model of a water molecule. The oxygen-hydrogen separation is $d = 9.57 \times 10^{-11}$ m. Each hydrogen atom has mass 1.0 u, and the oxygen atom has mass 16.0 u. Find the position of the center of mass.



General formalism



For the system of particles shown, the internal forces $\mathbf{f}_i$ between particles always occur in pairs with equal magnitude and opposite directions. Thus the **internal impulses sum to zero**.

The linear impulse and momentum equation for this system only includes the impulse of **external** forces.

$$\sum m_i(\vec{\mathbf{v}}_i)_1 + \sum \int_{t_1}^{t_2} \vec{\mathbf{F}}_i dt = \sum m_i(\vec{\mathbf{v}}_i)_2$$

Motion of the center of mass

For a system of particles, we can define a “fictitious” center of mass of an aggregate particle of mass m_{tot} , where m_{tot} is the sum ($\sum m_i$) of all the particles. This system of particles then has an aggregate velocity of $\mathbf{v}_G = (\sum m_i \mathbf{v}_i) / m_{\text{tot}}$.

The motion of this fictitious mass is based on motion of the center of mass for the system.

The position vector $\mathbf{r}_G = (\sum m_i \mathbf{r}_i) / m_{\text{tot}}$ describes the motion of the center of mass.

Conservation of linear momentum



When the **sum of external impulses** acting on a system of objects is **zero**, the linear impulse-momentum equation simplifies to

$$\sum m_i(\mathbf{v}_i)_1 = \sum m_i(\mathbf{v}_i)_2$$

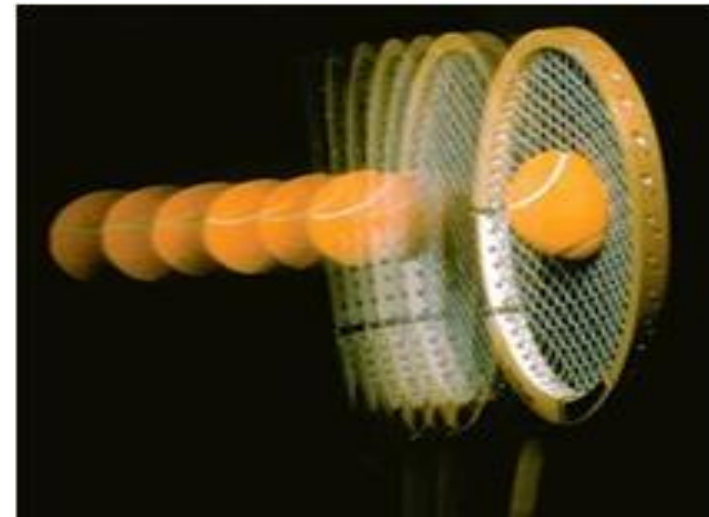
This equation is referred to as the **conservation of linear momentum**. Conservation of linear momentum is often applied when particles collide or interact. When particles impact, only **impulsive forces** cause a change of linear momentum.

The sledgehammer applies an impulsive force to the stake. The weight of the stake is considered negligible, or non-impulsive, as compared to the force of the sledgehammer. Also, provided the stake is driven into soft ground with little resistance, the impulse of the ground acting on the stake is considered non-impulsive.

Quiz

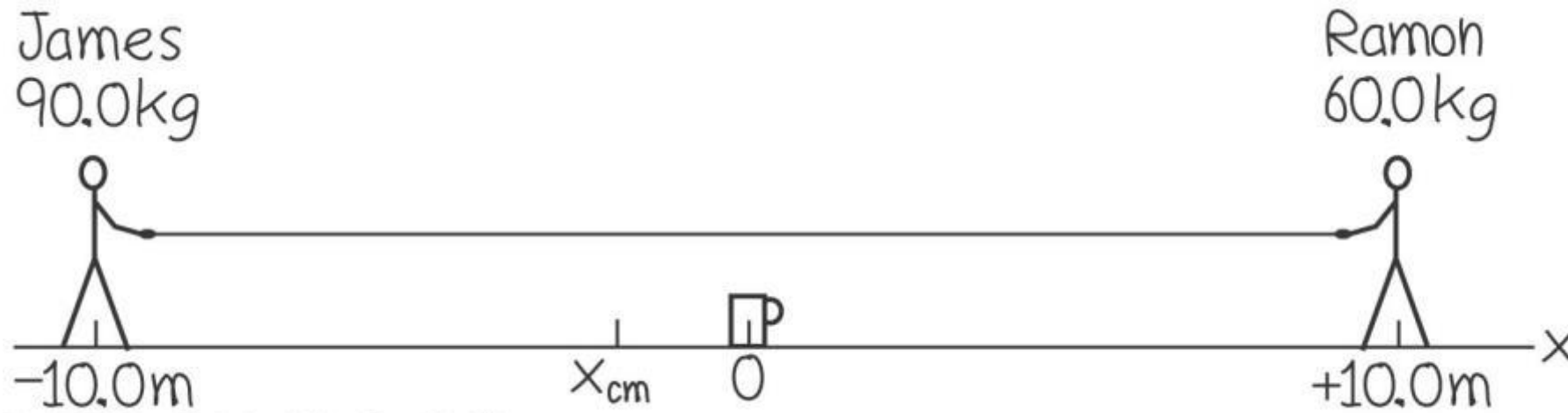
1. The internal impulses acting on a system of particles always _____.
A) equal the external impulses B) sum to zero
C) equal the impulse of weight D) None of the above.

2. If an impulse-momentum analysis is considered during **the very short time of interaction**, as shown in the picture, weight is a/an _____.
A) impulsive force
B) explosive force
C) non-impulsive force
D) internal force



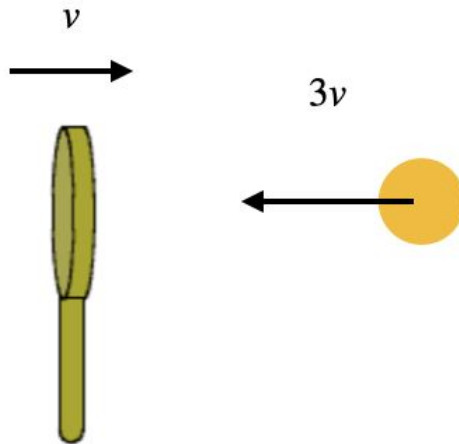
Example: Tug-of-war on the ice

- A tug-of-war occurs on frictionless ice. When James has moved 6.0 m toward the mug, how far and in what direction has Ramon moved?



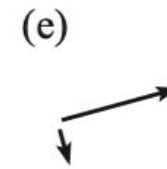
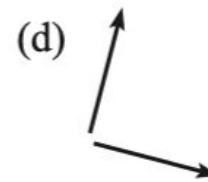
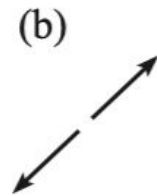
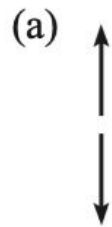
Example

A paddle hits a ping-pong ball. The paddle moves with speed v to the right, and the ping-pong ball moves with speed $3v$ to the left. The collision is elastic. What is the resulting speed of the ping-pong ball? (Assume that the paddle is much more massive than the ping-pong ball.)



Quiz

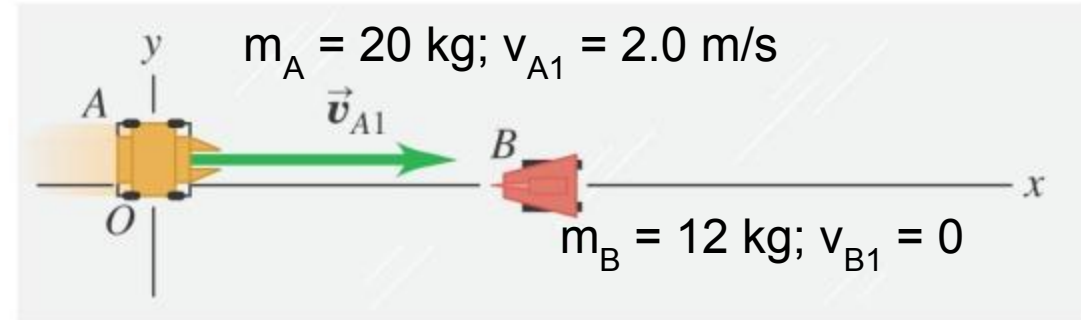
- A mass m moves with a given speed and collides (not necessarily head-on) elastically with another mass m that is initially at rest, as shown in Figure. Which of the figures shows a possible outcome for the two velocities?



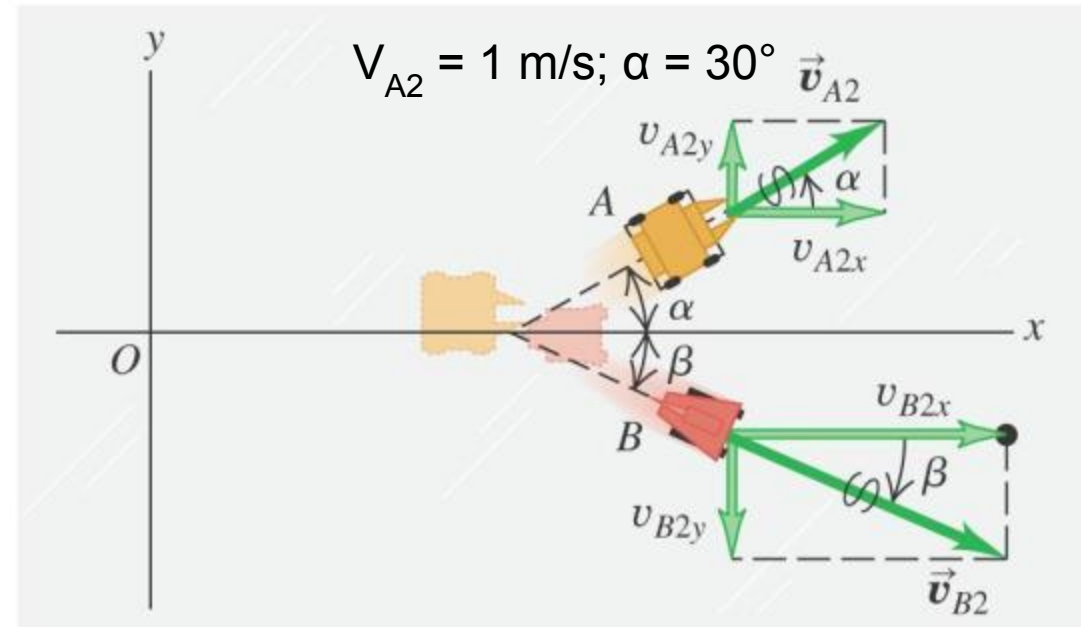
Example

- Two robots collide and go off at different angles.
- What is the final velocity of robot B?

(a) Before collision

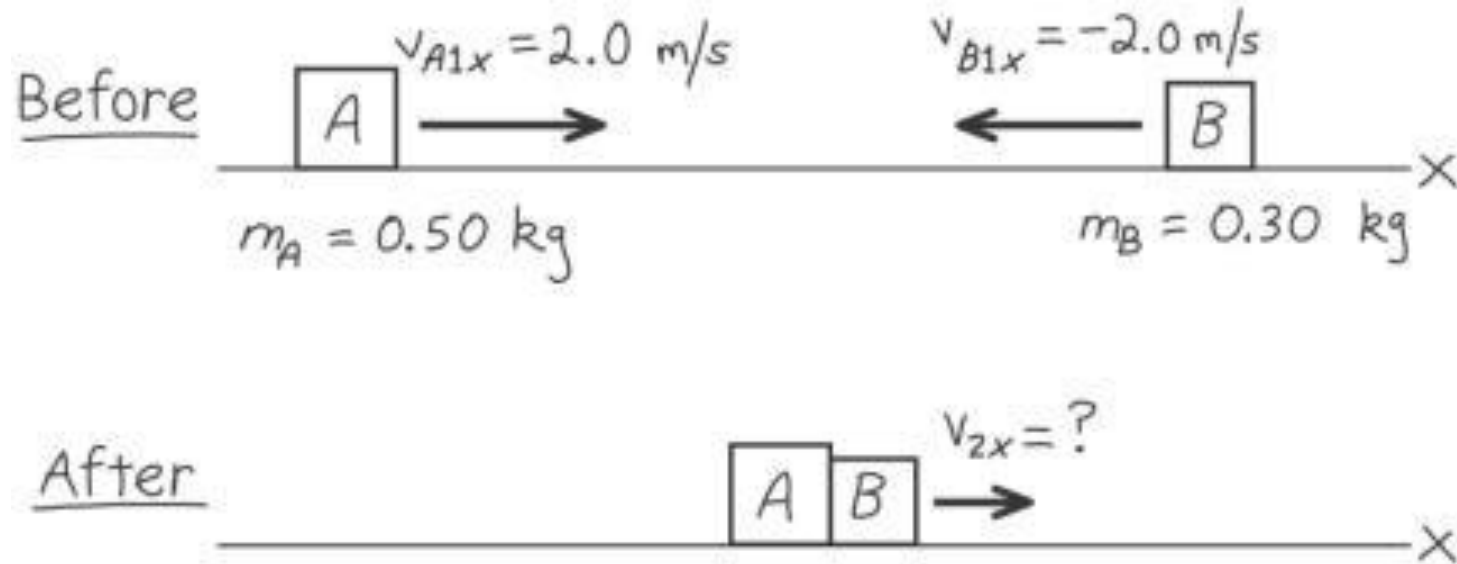


(b) After collision

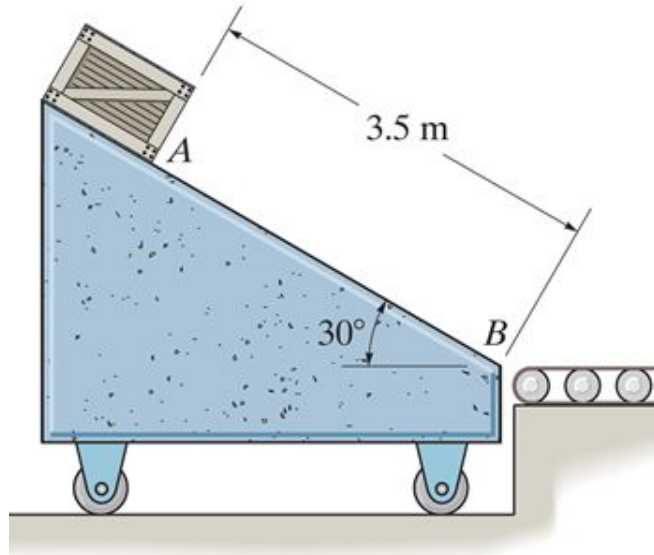


Example: Inelastic collision

- Follow the figure, which illustrates a completely inelastic collision. Find v_{2x} and change in kinetic energies of the system



Example

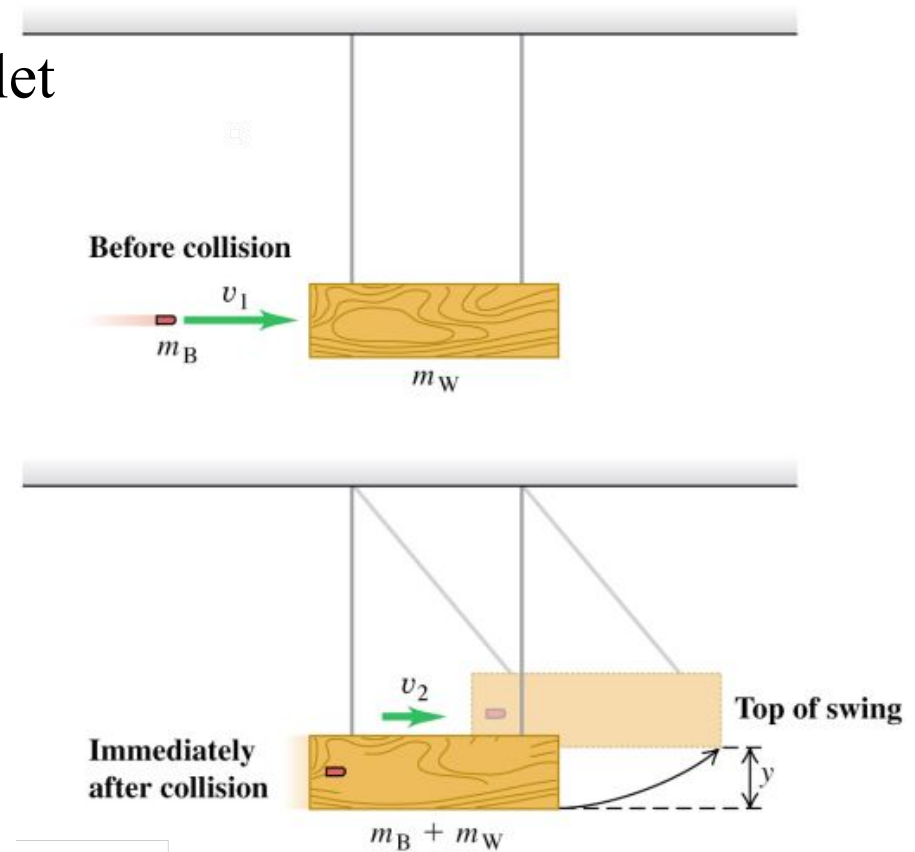


Given: The free-rolling ramp has a mass of 40 kg. The 10-kg crate slides from rest at A, 3.5 m down the ramp to B. Assume that the ramp is smooth, and neglect the mass of the wheels.

Find: The ramp's speed when the crate reaches B.

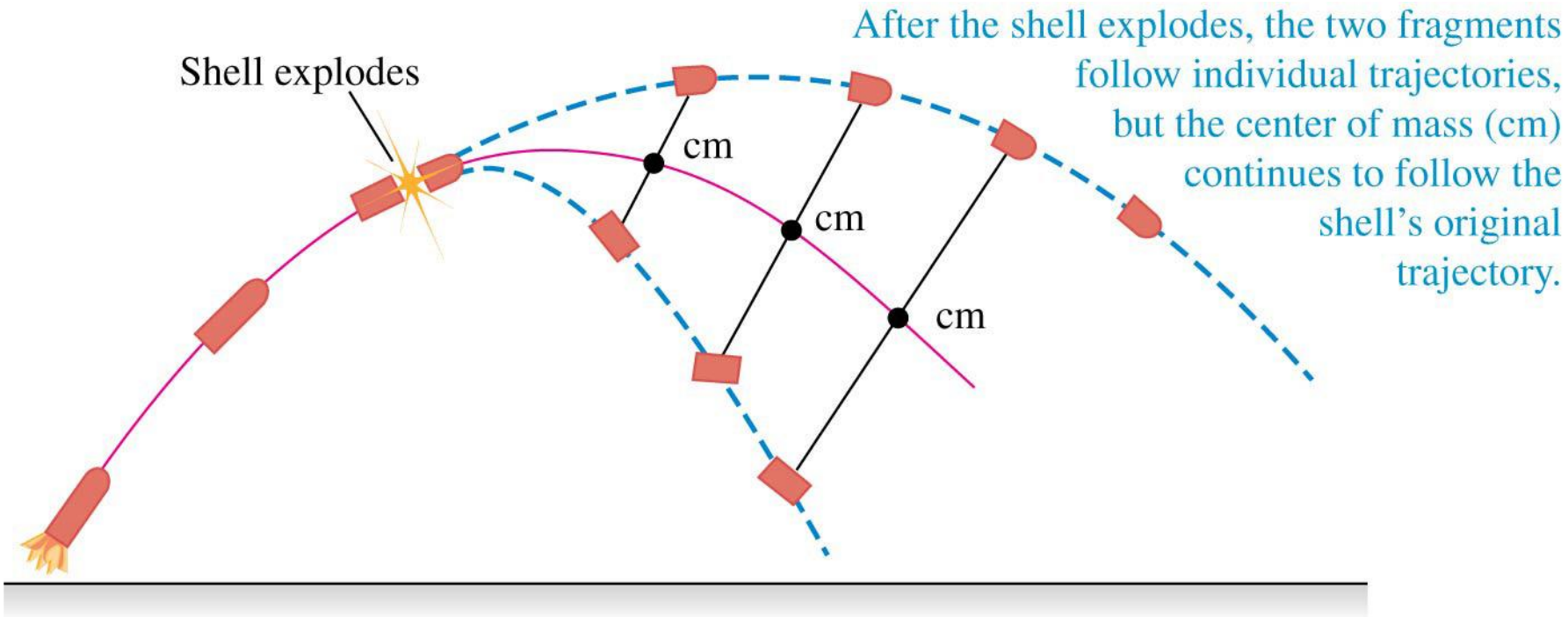
The ballistic pendulum

- Ballistic pendulums are used to measure bullet speeds.
- What is the initial speed v_1 of the bullet, in terms of the other variables?



External forces and center-of-mass motion

- When a body or a collection of particles is acted on by external forces, the center of mass moves as though all the mass were concentrated at that point and it were acted on by a net force equal to the sum of the external forces on the system.



Rocket propulsion

- Why can a rocket function in outer space where “it doesn’t have anything to push against”?



Rocket propulsion

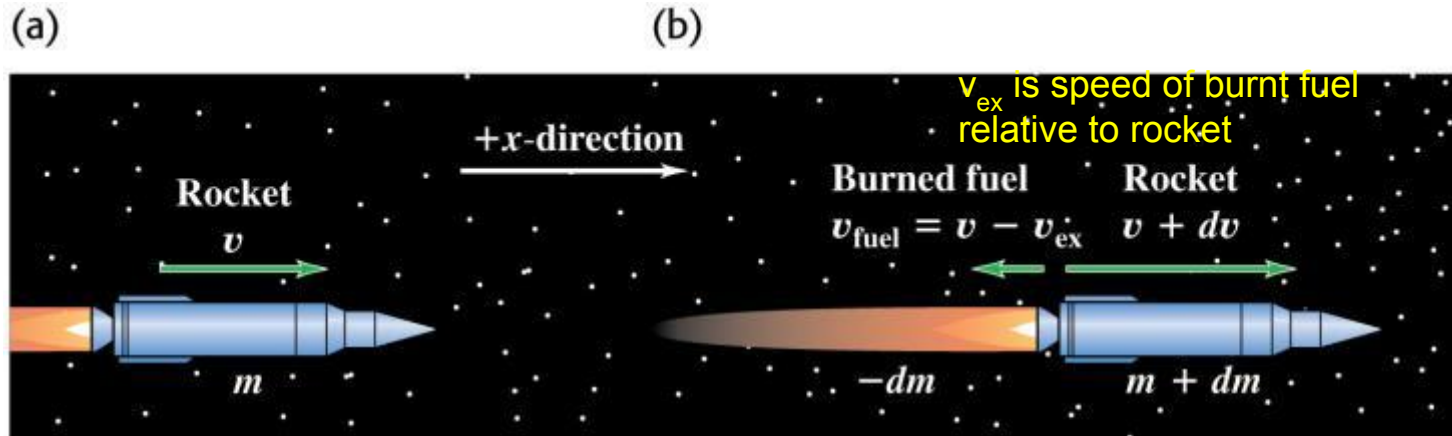
- As a rocket burns fuel, its mass decreases.
- To provide enough thrust to lift its payload into space, this *Atlas V* launch vehicle ejects more than 1000 kg of burned fuel per second at speeds of nearly 4000 m/s.



Rocket propulsion

- Initial total momentum = mv
- After dt , total momentum = $(m+dm)(v+dv)+(-dm)(v-v_{ex})$
- By conservation of momentum, $m dv = -dm v_{ex} - dm dv$
- Ignoring $dm dv$ and dividing by dt

$$m \frac{dv}{dt} = F = -v_{ex} \frac{dm}{dt}$$



At time t , the rocket has mass m and x -component of velocity v .

At time $t + dt$, the rocket has mass $m + dm$ (where dm is inherently *negative*) and x -component of velocity $v + dv$. The burnt fuel has x -component of velocity $v_{fuel} = v - v_{ex}$ and mass $-dm$. (The minus sign is needed to make $-dm$ *positive* because dm is negative.)

Rocket propulsion

- If the exhaust speed is constant, $m dv = -dm v_{ex}$

$$dv = -v_{ex} \frac{dm}{m}$$

$$\int_{v_0}^v dv = -v_{ex} \int_{m_0}^m \frac{dm}{m}$$

$$v = v_0 + v_{ex} \ln \frac{m_0}{m}$$

Tsiolkovsky rocket
equation in free space

- If $\ln(m_0/m) > 1$, the final speed of the rocket can be larger than v_{ex} .

Example

- Suppose 75% of the initial mass of the rocket is fuel, and v_{ex} is 2400 m/s, find the speed of the rocket, which is initially at rest, after all the fuel is consumed.

Summary

Linear Momentum

Impulse

Principle of linear momentum and impulse

Linear momentum conservation

Elastic collision

Inelastic collision

Center of mass and its motion

Rocket